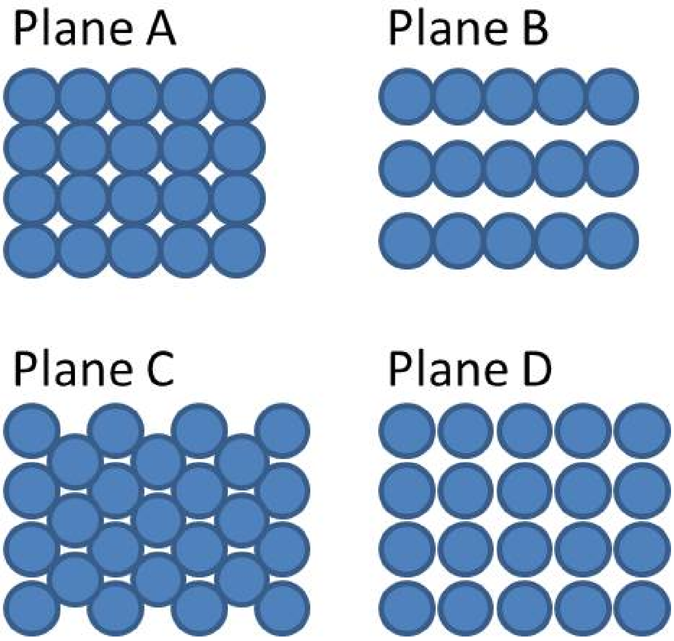
41680 F18 Written examination

Crystallography and crystal structures

Spørgsmål 1

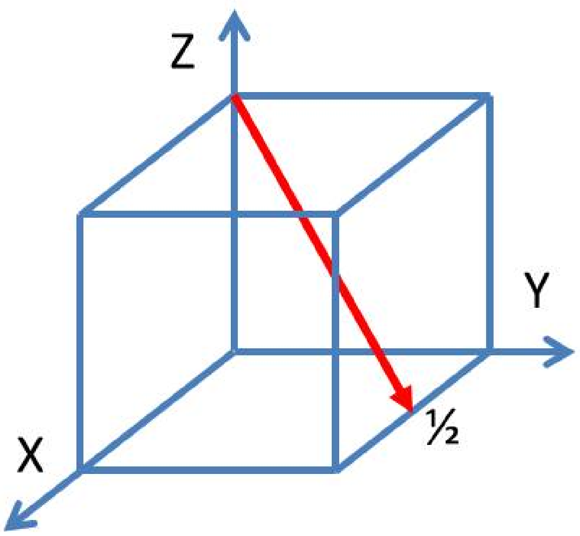
Which of the hard sphere models of crystallographic planes shown in the figure represents a {110} plane of a face-centered cubic lattice?



- [ ] Plane A
- [ ] Plane B
- [x] Plane C
- [ ] Plane D

Spørgsmål 2

Which is the correct crystallographic notation for the crystallographic direction shown in the figure in red?

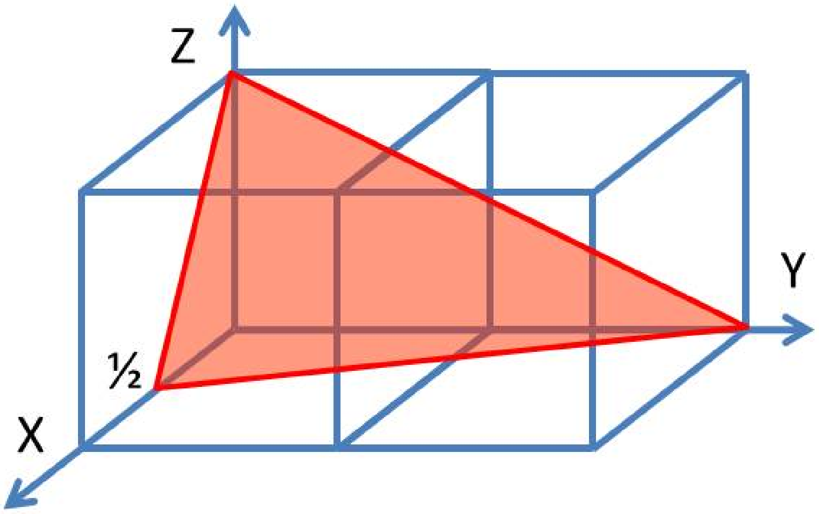


- [ ] $[211]$
- [ ] $[122]$
- [ ] $[21\bar{1}]$

- [x] $[12\overline{2}]$

Spørgsmål 3

Which is the correct crystallographic notation for the crystallographic plane shown in the figure in red?



- [ ] $\{124\}$
- [x] (142)
- [ ] (412)
- [ ] $\{421\}$

Spørgsmål 4

At low temperatures, Lithium has a face-centered cubic structure. What is its approximate density, if the molar mass of Lithium is 6.94 g/mol and its atomic radius is 0.152 nm?

- [ ] 0.41 g/cm^3
- [ ] 0.53 g/cm^3
- [x] 0.58 g/cm^3
- [ ] 0.78 g/cm^3

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Lattice defects

Spørgsmål 5

A specimen of pure aluminium is cooled slowly from room temperature to the temperature of liquid nitrogen (-196 °C). How does the defect content change?

- [x] the number of vacancies decreases
- [ ] the dislocation density decreases
- [ ] the number of grains decreases
- [ ] not at all

Spørgsmål 6

Which change in the defect content leads to an increased yield strength of an alloy compared to a pure metal?

- [ ] an increase in vacancy concentration
- [x] an increase in the phase boundary area
- [ ] an increase in grain size
- [ ] an increase in pore volume fraction

Spørgsmål 7

Dislocations are important for the mechanical behaviour of metals. Which of the the following statements is **not** justified?

- [ ] dislocations cause elastic stresses
- [ ] dislocations are carriers of plastic deformation
- [x] dislocations can act as obstacles against plastic deformation
- [ ] dislocations propagate fracture of metals

Spørgsmål 8

Increasing the dislocation density of a metal will increase ...

- [ ] ... the mass density
- [ ] ... Young's modulus
- [x] ... the yield strength
- [ ] ... the electrical conductivity

41680 F18 Written examination

Polymers

Spørgsmål 9

Polypropylen can exist in different configurations. Which configuration allows the highest crystallinity?

- [ ] the isotactic configuration
- [ ] the atactic configuration
- [x] the syndiotactic configuration
- [ ] the configuration has no influence on crystallinity

Spørgsmål 10

Polyethylen has quite different properties depending on its structure (amorphous or crystalline) as detailed in the table below. What is the expected Youngs's modulus for semi-crystalline polyethylen with a density of 0.920 g/cm^3?

	density (g/cm^3)	Young's modulus (MPa)
amorphous	0.870	60
crystalline	0.998	500

- [x] less than 90 MPa
- [ ] between 90 MPa and 240 MPa
- [ ] between 240 MPa and 400 MPa
- [ ] above 400 MPa

Spørgsmål 11

Semi-crystalline polymers have a peculiar mechanical behavior as ...

- [ ] ... they possess extremly low Young's moduli
- [ ] ... they do not develop necking and show viscous flow
- [ ] ... their flow curve shows a decrease in flow stress followed by work-hardening
- [x] ... they are brittle and fracture without plastic deformation

Spørgsmål 12

Two specimens (one made of polypropylene PP, the other made of polyvinylchloride PVC) have the same degree of polymerization. Which statement about their molar mass and melting temperature is correct?

- [ ] PP has both a higher molar mass and a higher melting temperature than PVC
- [ ] PP has a higher molar mass, but a lower melting temperature than PVC
- [x] PVC has both a higher molar mass and a higher melting temperature than PP
- [ ] PVC has a higher molar mass, but a lower melting temperature than PP

Spørgsmål 13

The glass transition temperature of a polymer is lowest, if ...

- [ ] the monomers have high molar mass
- [x] there are no double bonds in the main chain
- [ ] there are large side groups
- [ ] there are many side branches

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Ceramics

The table summarizes a few essential properties of cations and anions.

Element	Valence	Electro-nega-tivity	Ionic radius nm	Element	Valence	Electro-nega-tivity	Ionic radius nm
Li	+1	1.0	0.073	F	-1	4.1	0.133
Na	+1	1.0	0.102	Cl	-1	2.9	0.181
K	+1	0.9	0.138	Br	-1	2.8	0.196
Cs	+1	0.9	0.170	I	-1	2.2	0.220
Tl	+1	1.5	0.160	O	-2	3.5	0.140
Mg	+2	1.3	0.072	S	-2	2.4	0.184
Al	+3	1.5	0.053				
Si	+4	1.8	0.040				

Spørgsmål 14

Which of the following compounds has the smallest fraction of ionic bonding?

- [ ] Li_2S
- [ ] MgS
- [ ] Al_2O_3
- [x] SiO_2

Spørgsmål 15

Which of the following compounds has the same structure as CsI (cesium iodide)?

- [ ] LiF
- [x] TlCl
- [ ] NaBr
- [ ] KI

Spørgsmål 16

Why are ionic ceramics brittle?

- [ ] no dislocations exist at all
- [ ] existing dislocations cannot move
- [ ] dislocation storage leads to etremely high work-hardening
- [x] dislocation motion may result in neighboring negative charges

Spørgsmål 17

When the volume fraction of pores in a ceramic material is decreased (e.g. during manufacturing), ...

- [ ] the mass density decreases
- [x] Young's modulus increases
- [ ] the fracture strength decreases
- thermal insulation becomes better

☐

Spørgsmål 18

Which defects **cannot** be found in amorphous ceramics?

- [x] substitutional atoms
- [ ] dislocations
- [ ] pores
- [ ] cracks

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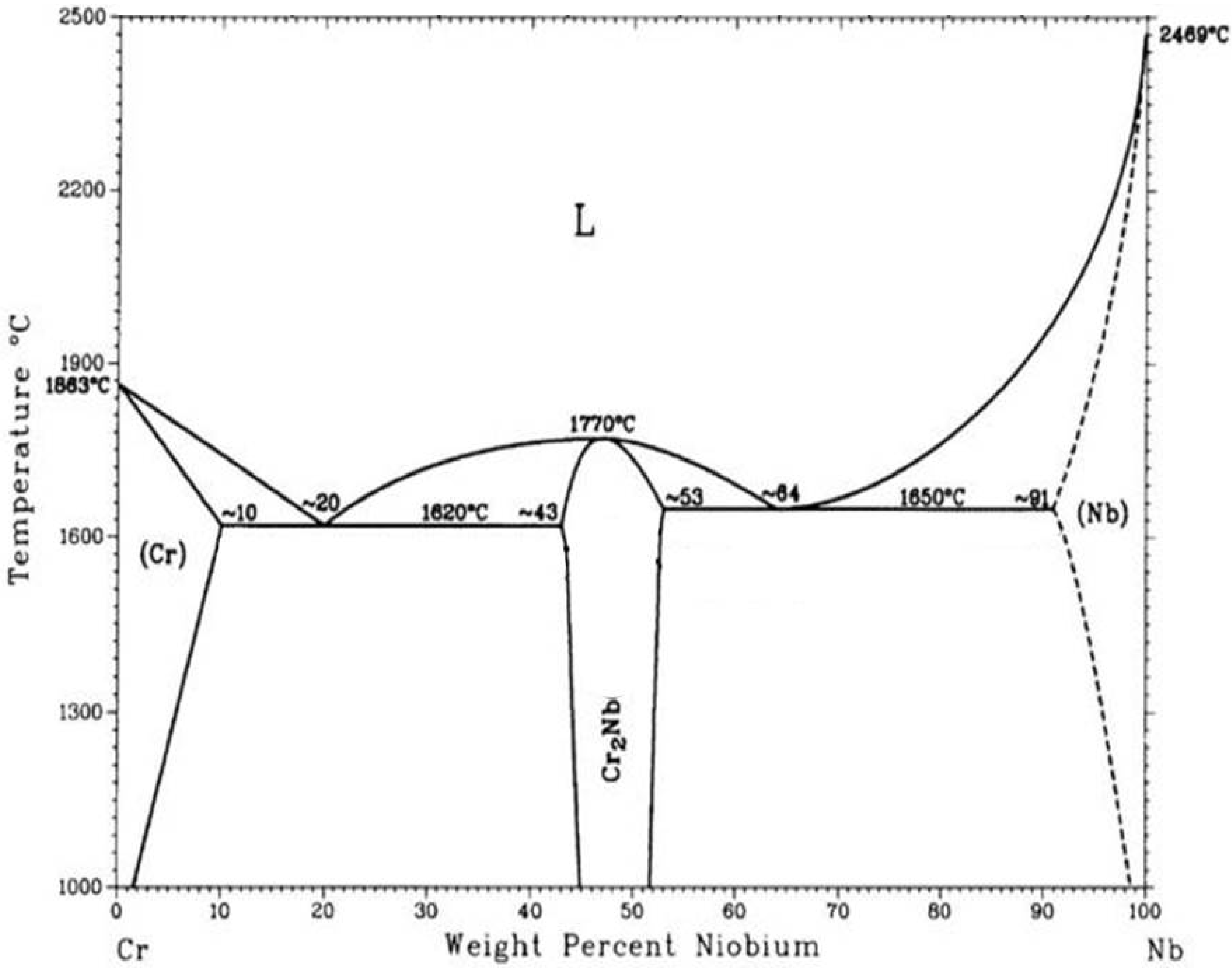
Phase diagram (I)

A melt of a chromium-niobium alloy containing 27 weight% Nb is slowly cooled from 2500 °C under conditions of thermal equilibrium.

The figure shows the (simplified) phase diagram of the binary system chromium-niobium (Cr-Nb). Dashed lines should be considered as full lines.



Spørgsmål 19

At which temperature does solidification of the chromium-niobium alloy with 27 weight% Nb start?

- [ ] 1620 °C
- [x] 1700 °C
- [ ] 1770 °C
- [ ] 1863 °C

Spørgsmål 20

Which is the first solid phase formed during the solidification of the chromium-niobium alloy with 27 weight% Nb?

- [x] (Cr)
- [ ] (Cr) + L
- [ ] Cr_2Nb
- [ ] Cr_2Nb + L

Spørgsmål 21

What is the maximal Nb content in the melt during the solidification process of the chromium-niobium alloy containing 27 weight% Nb ?

- [x] 20 weight% Nb

- [ ] 27 weight% Nb
- [ ] 47 weight% Nb
- [ ] 64 weight% Nb

Spørgsmål 22

Which solid phases are present after cooling of the chromium-niobium alloy containing 27 weight% Nb to 1000 °C?

- [ ] (Cr) with 10 weight% Nb and Cr_2Nb with 43 weight% Nb
- [x] (Cr) with 1.5 weight% Nb and Cr_2Nb with 45 weight% Nb
- [ ] a eutectic with 20 weight% Nb and Cr_2Nb with 45 weight% Nb
- [ ] a eutectic with 20 weight% Nb and Cr_2Nb with 53 weight% Nb

Spørgsmål 23

What is the mass of Cr_2Nb in 50 kg of the chromium-niobium alloy with 27 weight% Nb after cooling to 1000 °C?

- [ ] 13.5 kg
- [x] 20.7 kg
- [ ] 29.3 kg
- [ ] 36.5 kg

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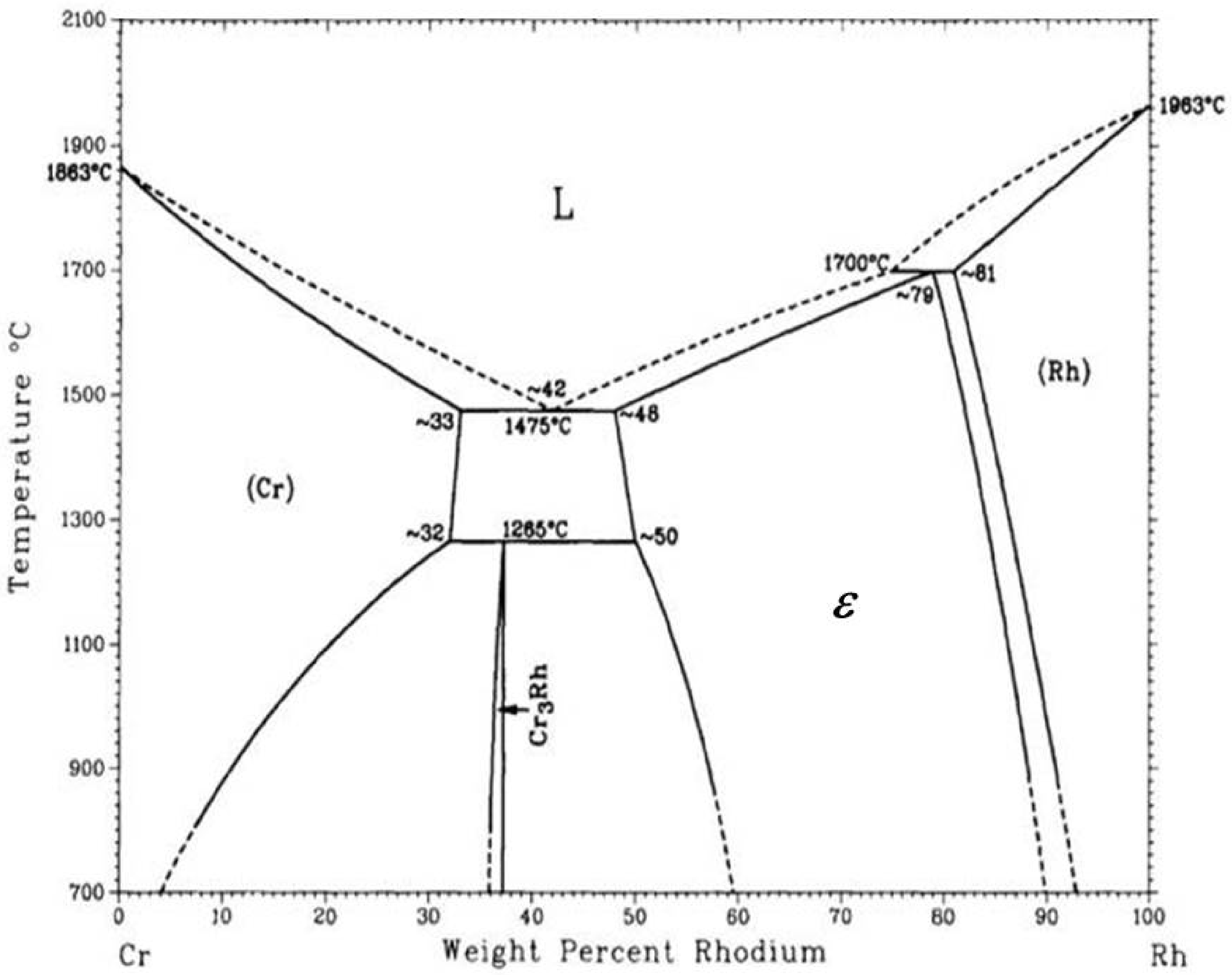
Phase diagram (II)

The figure shows the phase diagram of the binary system chromium-rhenium (Cr-Rh). Dashed lines should be considered as full lines.



Spørgsmål 24

Consider chromium-rhodium alloys of all possible compositions at 1400 °C. Where can Cr atoms **not** be found in thermal equilibrium?

- [ ] on regular lattice sites in (Cr) solid solution
- [ ] on interstitial sites in (Cr) solid solution
- [ ] in the ε phase (epsilon phase)
- [x] as substitutional atoms in the (Rh) solid solution

Spørgsmål 25

Consider cooling of a chromium-rhodium alloy containing 42 weight% Rh from 2000 °C to 700 °C. Which statement is correct?

- [ ] (Cr) will not be formed at any instant during the cooling process.
- [x] (Rh) will not be formed at any instant during the cooling process.
- [ ] Neither (Cr), nor (Re) will be formed at any instant during the cooling process.
- [ ] All four solid phases will be formed at some stage during the cooling process.

Spørgsmål 26

Which microstructure is expected, if a melt of a chromium-rhodium alloy containing 42 weight% Rh is slowly cooled to 700 °C?

- [ ] Solely an eutectic structure with alternating (Cr) and ε lamellae.
- [ ] A microstructure consisting of the two different phases (Cr) and ε.
- [x] A microstructure consisting of the two different phases Cr_3Rh and ε.
- [ ] A microstructure consisting of three different phases: (Cr), Cr_3Rh and ε.

Spørgsmål 27

What is the main strengthening effect when pure rhodium is alloyed with 5 weight% chromium?

- [ ] solid solution strengthening in (Cr) solid solution
- [ ] particle strengthening by Cr_3Rh particles
- [ ] boundary strengthening by phase boundaries with ε phase
- [x] solid solution strengthening in (Rh) solid solution

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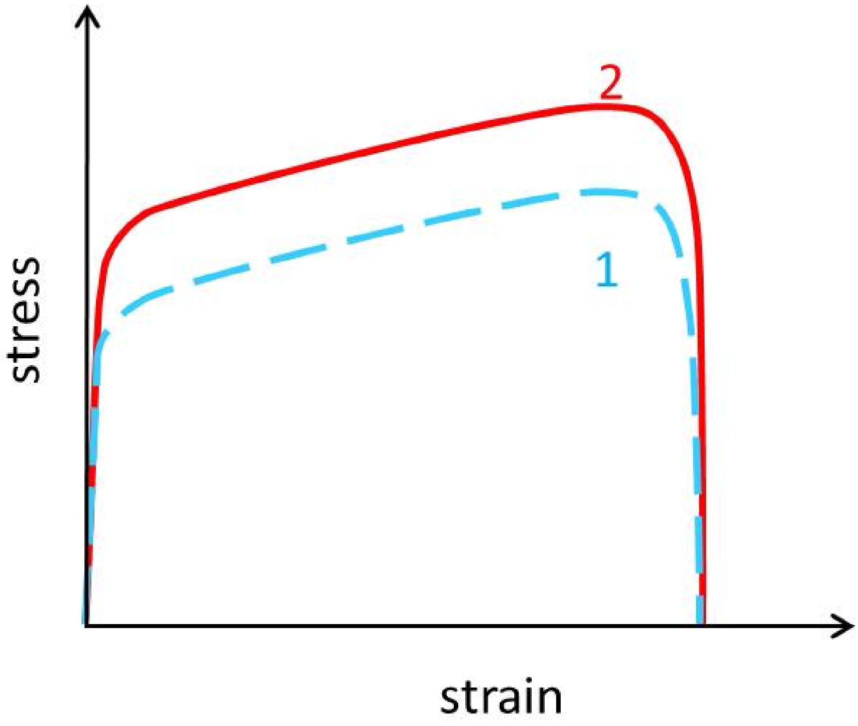
Mechanical behavior

The stress strain diagramm shows two flow curves obtained for two different metallic specimens (1 and 2) at room temperature.



Spørgsmål 28

Compare tensile strength and resilience of the two metallic specimens.

- [ ] 1 has both a higher tensile strength and a higher resilience than 2.
- [ ] 1 has a higher tensile strength, but a lower resilience than 2.
- [ ] 2 has a higher tensile strength, but a lower resilience than 1.
- [x] 2 has both a higher tensile strength and a higher resilience than 1.

Spørgsmål 29

What could be the difference between these two metallic specimens?

- [ ] Both specimens could have the same composition, one work-hardened, the other not work-hardened.
- [x] Both specimens could have the same composition, one with smaller grain size, the other with larger grains.
- [ ] Both specimens could have the same composition, one with smaller grain size and work-hardened, the other with larger grain size, but not work-hardened.
- [ ] The two specimens must obviously have a different chemical composition.

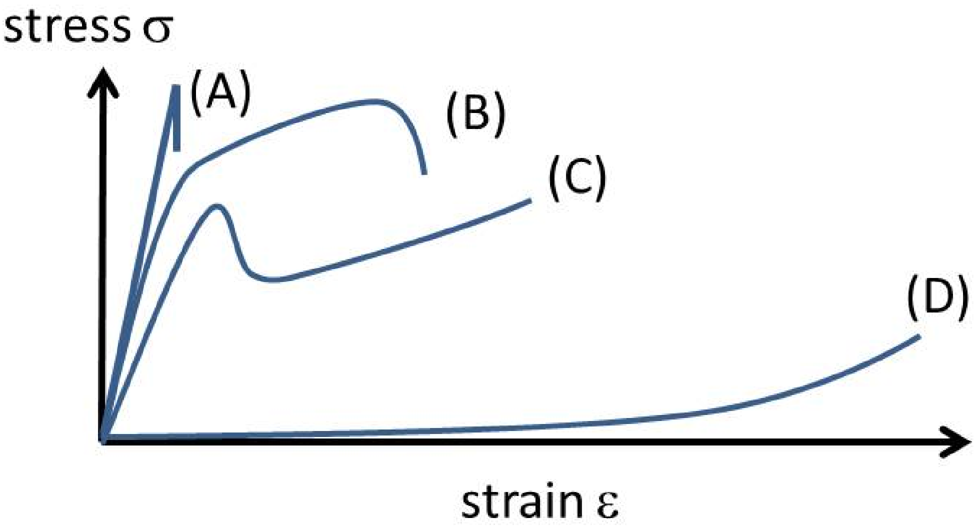
Spørgsmål 30

Which of the two metallic specimens should be selected for producing a light and stiff pillar of a given length?

- [ ] Assuming the same mass density for both specimens, specimen 1.
- [ ] Assuming the same mass density for both specimens, specimen 2.
- [x] Assuming the same mass density for both specimens, both specimens are equally well suited.
- [ ] The specimen with the lower mass density, independent of the flow curve.

Spørgsmål 31

Which flow curve is characteristic for an elastomer?



- [ ] A
- [ ] B
- [ ] C
- [x] D

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Image analysis and materials modelling

Spørgsmål 32

A median filter is a nonlinear filter that uses a window of e.g. 5-by-5 pixels. A pixel value in the filtered image is obtained by ...

- [ ] ... summing the elements in the window around a pixel and dividing the sum by the number of elements in the window.
- [x] ... sorting the values of the pixels in the window and selecting the central element (i.e. for a 5-by-5 filter the 13th largest value).
- [ ] ... finding the average of the pixels values in the window and selecting the pixel value closest to this value (using Eucledian distance).
- [ ] ... computing the difference in pixel value between the pixel to the right and the pixel to the left of the pixel in question.

Spørgsmål 33

A simple pixel classification can be obtained using a threshold. The threshold classifier is given by: k=0 if I>t, else k=1, where k is a class label, t is a threshold value and I is the pixel value. What is the sum of the pixel values in the thresholded image, if one applies the classification rule with a threshold of t=100 on the image below.

91	87	88	92	99
93	87	90	97	100
90	89	106	110	97
94	90	93	101	95
92	98	105	117	107
87	82	84	92	108

- [ ] 0
- [ ] 6
- [ ] 7
- [x] 23

Spørgsmål 34

Given a volumetric image of size 300-by-300-by-600 voxels, which of the following statements is true?

- [ ] The volume contains 27 million voxels in total and can be viewed as six hundred 300-by-300 image slices in the x-y plane.
- [x] The volume contains 54 million voxels in total and can be viewed as three hundred 300-by-600 image slices in the y-z plane.
- [ ] The volume contains 54 million voxels in total and each voxel can only be represented by 8 bit.
- [ ] The volume contains 600 independent images stacked on top of each other and can only be analyzed in 2D.

Spørgsmål 35

Which statement about Fick's second law of diffusion is correct?

- [ ] According to Fick's second law of diffusion, the higher the diffusion coefficient, the lower the diffusional flux.
- [ ] According to Fick's second law of diffusion, a diffusional flux will always be constant over time.
- [x] Fick's second law of diffusion describes the change of concentration of a diffusing species with time.
- [ ] Fick's second law of diffusion is only applicable for steady state systems.

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Transport phenomena

Spørgsmål 36

A metallic wire with a cylindrical cross section of diameter 1 mm and a length of 100 mm is observed to transport a current of 340 mA when it is subjected to a voltage difference of 1 mV. Which material is consistent with this observation?

Metal	Fe	Au	Pt	Stainless steel
Electrical conductivity $(\Omega\ m)^{-1}$	$1.0\cdot10^7$	$4.3\cdot10^7$	$0.94\cdot10^7$	$0.2\cdot10^7$

- [ ] Fe
- [x] Au
- [ ] Pt
- [ ] Stainless steel

Spørgsmål 37

An aluminium wire with a cross section of $1.6\cdot10^{-6}$ m^2 carries an electrical current of 2 A. What is the drift speed of the electrons in the wire, assuming that each Al atom contributes with a single free electron? The mass density of aluminium is 2.7 g/cm^3 and its molecular weight 27 g/mol.

- [x] 0.13 mm/s
- [ ] 32 cm/s
- [ ] 44 m/s
- [ ] 7.7 km/s

Spørgsmål 38

Consider the intrinsic semiconductor GaAs at room temperature. What type of carriers are active in conducting an electrical current?

- [x] Electrons near the conduction band
- [ ] Electrons near the valence band
- [ ] Holes in the conduction band
- [ ] Holes in the valence band

Spørgsmål 39

Why does polycrystalline pure nickel (Ni) have a much higher thermal conductivity than silica (SiO_2) glass?

- [ ] Ni is denser than SiO_2.
- [ ] Ni is magnetic, SiO_2 not.
- [x] Ni has metallic bonds with free electrons.
- [ ] Ni has an abundance of lattice defects.

Spørgsmål 40

A hollow cylinder with a total surface area of 500 m^2 is considered as a simplifed model for an airplane. The wall has a thickness of 1 cm everywhere and is made of a material with a thermal conductivity of 0.2 W/(K m). The temperature inside the cylinder is 25 °C and outside -50 °C. What is the total energy loss during a 10 h flight assuming a steady state heat flow?

- [ ] 340 kJ
- [x] 12 MJ
- [ ] 82 MJ
- [ ] 27 GJ